

VYUHA

Agentic OODA Mesh & ST-GNN for Proactive Fraud Interception

PROBLEM STATEMENT & TEAM DETAILS

TEAM NAME: Vyuha

PROBLEM STATEMENT: Vyuha: Agentic OODA Mesh & ST-GNN for Proactive Fraud Interception

S.No	NAME	GENDER	AREA OF EXPERTISE
1	Trombokendu Chakraborty	Male	AI Architecture, LangGraph, Backend
2	Alaukik Saxena	Male	UI/UX Design, HCI, Frontend Friction
3	[Name]	[Gender]	[Expertise]
4	[Name]	[Gender]	[Expertise]

PROPOSED SOLUTION

Core Problem: Reactive systems fail against psychological coercion and real-time social engineering.

The Vyuha Solution: On-device Edge-AI SDK acting as an intelligent, contextual behavioral firewall.

The Metric: Targets the **₹1,935+ Crore** lost to digital arrests (2024).

The Mechanism: Contextual UI friction actively breaks the scammer's isolation loop **before** the UPI PIN is ever entered.

 Conceptual Flowchart of Vyuha Solution

OUTLINE OF UNIQUE & INNOVATIVE SOLUTION

Zero-Drain Triggers

Listens to Android OS broadcasts (Call > 20 mins + UPI App active).

Zero battery drain. No continuous telemetry required.

Agentic Friction

MoE Router ensures local processing with <

50ms latency.

95% of safe traffic is bypassed instantly.

Trusted Node

Breaks psychological isolation effectively.
Family emergency ping deployed instantly to verified contacts.

 Tech Icons representing Microchip, Router, and Shield Nodes

TECHNICAL APPROACH

Observe: Passive OS Event Listeners capture environmental context.

Orient: Local ST-GNN rapidly checks receiver mule-ring probability.

Decide: Contextual RL Bandit dynamically selects the appropriate friction level.

Act: Local UI Overlay deployed + Global Graph Memory updated securely.

 Detailed LangGraph Multi-Agent Mesh Architecture Diagram

FEASIBILITY & VIABILITY

Integration: Seamless B2B Edge-AI SDK for major gateways (Razorpay/PayU). Bypasses rigid, closed NPCI sandboxes.

Execution: Highly quantized models run entirely locally on the user's device for speed and privacy.

Compliance: 100% DPDP Act compliant. Zero cloud PII transfer guarantees user data sovereignty.

 Spatio-Temporal Graph Neural Network Visualization

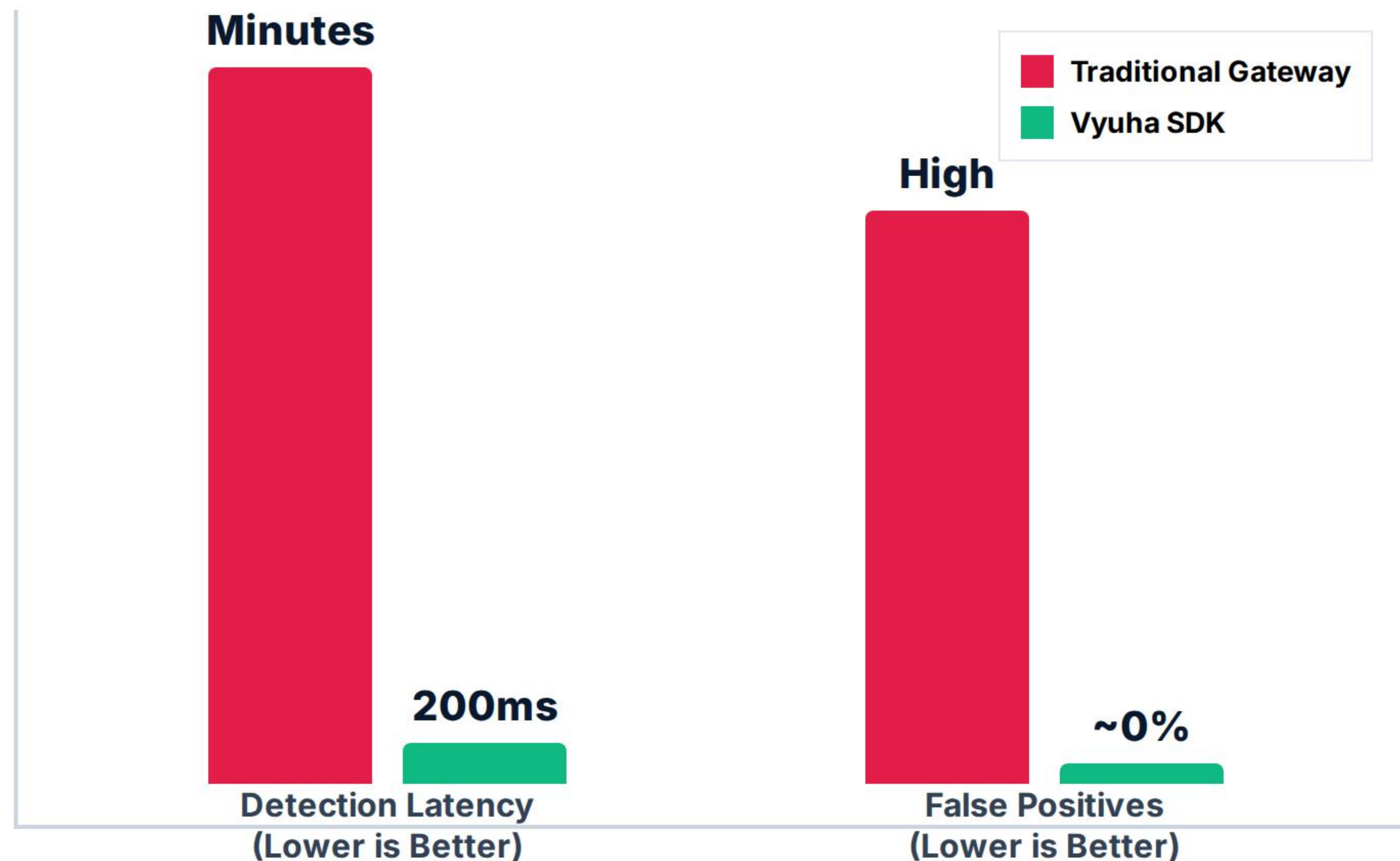
IMPACT & BENEFITS

Current State: 100% reliance on reactive, post-fraud chargebacks which often fail to recover funds.

Vyuha Impact: Drastically shifts security upstream to the exact point-of-intent.

User Benefit: Protects the most vulnerable digital demographics via intelligent, non-intrusive UI interventions.

Fraud Prevention Efficacy



BUSINESS MODEL

Target Market

B2B SaaS primarily engineered for Payment Gateways & Sponsor Banks.

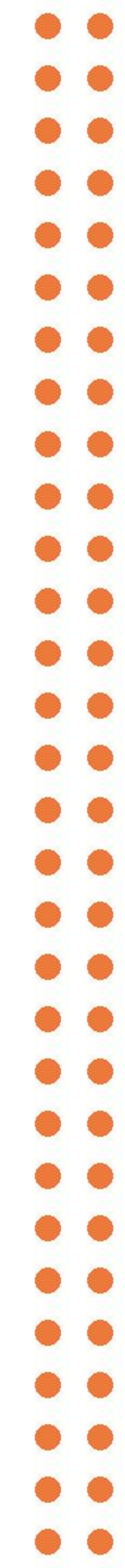
Revenue Streams

Volume-based API licensing + Premium tiers for advanced ST-GNN compute.

ROI Generation

Effectively eradicates manual chargeback OPEX and regulatory non-compliance fines.





RESEARCH & REFERENCES

- **Architecture Validation:** rigorously validated against current operational gaps found in RBI's MuleHunter.AI and Razorpay Vulcan.
- **RL Friction Models:** Directly grounded in current contextual bandit personalization literature, optimizing for adaptive UI thresholding without user fatigue.
- **Spatio-Temporal Graph Logic:** Utilizes current state-of-the-art anomaly camouflage mitigation techniques specifically designed for high-density transactional networks.
- **Scale & Capability:** Architected to intercept a significant portion of the **₹11,158 Cr** saved manually by I4C in 2026 through automated, localized intervention.